

VIDEO SWITCHER

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With substantial increase in number of thefts, surveillance systems for stores, offices and homes have become very important today. Usually, a system with a single camera and a TV with video input is sufficient for surveillance of a small area. But if multiple cameras need to be hooked up to a single display unit (such as TV), a video switcher is required to select multiple video inputs. The video switcher switches automatically between several camera inputs in sequence. Unfortunately, the cost of such video switchers available in the market is too high. Here we have described an affordable and easy-to-build video switcher.

PARTS LIST

Semiconductors:

IC1	- 7812, 12V regulator
IC2	- NE555 timer
IC3	- CD4017 decade counter
IC4	- CD4066 quad bilateral switch
T1, T2	- BFW16A npn transistor
D1-D4	- 1N4007 rectifier diode
LED1-LED4	- 5mm LED

Resistors (all 1/4-watt, 5% carbon unless stated otherwise):

R1, R2, R6-R15	- 150-ohm
R3	- 820-ohm
R4	- 680-ohm
R5	- 100-ohm
R16-R19	- 1-kilo-ohm
R20	- 22-kilo-ohm
R21, R22	- 10-kilo-ohm
VR1	- 680-kilo-ohm potentiometer

Capacitors:

C1	- 1000µF, 50V electrolytic
C2	- 0.1µF ceramic disk
C3, C4	- 2.2nF ceramic disk
C5-C8	- 0.47µF, 35V electrolytic
C9	- 22µF, 35V electrolytic
C10	- 0.01µF ceramic disk

Miscellaneous:

X1	- 230V AC primary to 15V, 500mA secondary transformer
DIP Sw1 (S1-S4)	- 4-way DIP switch
S5	- SPST on/off switch
	- BNC connector

Fig. 1 shows the block diagram of video switcher. This video switcher displays up to four camera inputs on a single monitor and the number of inputs is selected by a selection switch on the circuit board. The camera selection feature avoids any blank display if less than four cameras are used. The display sequence can be generated by a sequencing counter.

When the signal provider sends one video signal out of four (in a sequence) to the video amplifier for television display, an oscillator provides clock signal to the sequence counter. In normal run mode, the video signals from cameras are switched at a rate that can be varied with a panel-mounted control (scan rate). The switching rate can be set from about once per second

bilateral switches, which work in conjunction with the decade counter. The output of the decade counter is also used to light a series of LEDs to indicate which camera's video signal is on display. Video cameras are connected to the video input terminal. The inputs are terminated with 75Ω resistors (two 150Ω resistor connected in parallel) to prevent cable reflections.

The sequencing counter is built around CD4017. It produces one positive output pulse at a time on each of its ten outputs in sequence for every clock pulse. The first four outputs Q0 through Q3 drive LED1 through LED4 with current limiting resistor R16 through R19, respectively. The outputs Q0 through Q3 of decade counter CD4017 drive the four bilateral

switch of CD4066

to provide video signal to the amplifier input. When Q0 output of IC3 goes high, control pin 12 of bilateral switch IC4(A) connects signal from video1 to the amplifier input. This available signal is amplified by a very simple am-

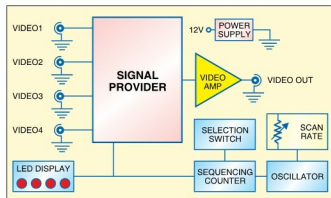


Fig. 1: Block diagram of video switcher

plifier made with transistors T1 and T2. Transistor T1 performs the job of amplification. Transistor T2 is wired as an emitter follower to increase current gain. This amplifier circuit is most suitable for cable TV systems using 75Ω coaxial cables and works fine up to 150 MHz.

The number of cameras connected to the video switcher is set with 4-way DIP switch SW1 (four switches inside, say S1, S2, S3, S4). Each switch (S1 through S4) in DIP switch is connected to outputs Q1 through Q4 of

Circuit description. The heart of the video switcher is a decade counter CD 4017 (IC3). Video inputs are selected by

